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CTRL "C" – CTRL "V"; using gaming peripherals to improve library workflows and enhance staff efficiency

Ryan Litsey, Rea Harris, and Jessie London

Texas Tech University, Lubbock, TX

ABSTRACT
Library workflows are an area where repetitive stress can potentially reduce staff efficiency. Day to day activities that require a repetitive motion can bring about physical and psychological fatigue. For library managers, it is important to seek ways in which this type of repetitive stress can be alleviated while having the added benefit of increasing staff productivity and efficiency. Another area where repetitive stress plays a role is in the computer gaming arena. The tasks undertaken by a computer gamer are not unlike the tasks undertaken by a library staff member. It is in this arena of computer gaming that libraries can find new tools and methods for reducing repetitive stress, computer based stress, as well an increase efficiency and effectiveness. This type of improvement can be accomplished by incorporating gaming peripherals into the workflow of the library staff member.

KEYWORDS
Gaming peripherals; ergonomics; interlibrary loan; staff improvement; flow

Introduction

Libraries can be a bastion of repetitive stress tasks. In some respect that is part of the library world. Thousands of books need to be cataloged a certain way and check out processes happen numerous times over the course of a library shift. These types of routine processes can create stress both on the body and the psyche. Many library staff has developed different ways of confronting the different side-effects of these repetitive tasks. Some are specific hand positioning when doing tasks to relive stress; another could be the use of a brace or specialized workspaces. These types of ergonomic solutions have been addressed repeatedly not only in libraries, but also in general in the industry. There is, however, another area that can work in conjunction with ergonomic changes that can also help alleviate stress, increase productivity and improve staff task accuracy. This new area is in the use of peripherals that are designed to reduce stress and increase accuracy for another segment of society that utilizes repetitive tasks.

The computer gaming community spends countless hours performing the same mouse and keyboard functions. This group places a premium on accuracy and

CONTACT Ryan Litsey ryan.litsey@ttu.edu Texas Tech University, PO Box 40002, Lubbock, TX, 79409, USA.
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speed. There is a large market for computer peripherals that can help gamers meet their needs. These peripherals are an untapped market for libraries. By incorporating gaming peripherals into the library staff day to day functions, libraries can increase speed, accuracy and relieve the stress of repetitive tasks on library employees. Additionally, reducing the psychological and physical strain of processing requests allows library staff to be in the work "flow" for longer periods of time. This leads to greater job satisfaction and better work performance.

To understand how gaming peripherals can be integrated into the library it is important to think of the repetitive strain as a three-part issue. (MacKinnon & Novak, 1997) The first part is the amount of tissue damage that occurs as part of the assigned task. Second, are the individual parameters: how well conditioned is the employee to perform the task? The third is hard to quantify and is where the focus of this paper can provide real assistance, which is in the area of the psychosocial factors. To adequately relieve Repetitive Stress/Strain Injuries (RSI) libraries will need to address all three.

The literature on ergonomics and repetitive stress in libraries and the workplace tends to focus on the first two criteria. However, to provide an adequate solution, the psychosocial aspects must be addressed. This is where gaming peripherals can relive all aspects that contribute to repetitive stress tasks. To understand how gaming peripherals can offer a comprehensive solution it is necessary to first understand repetitive stress and the role it plays in the modern library workplace. Second, the overlooked importance of the psychosocial understood by the concept of "flow". Finally, this research will demonstrate how gaming peripherals can offer a comprehensive solution to the problem of repetitive stress in libraries.

Repetitive stress/strain

Repetitive Stress/Strain Injuries (RSI) have received a lot of focus in the age of computers. RSI is defined by Byl (2006) as, "cumulative trauma disorders, repetitive motion injuries, and work-related musculoskeletal disorders are synonymous terms describing a broad range of injuries resulting from excessive use of the spine, arm, hand, leg, or foot." (Byl, 2006, pp.1382). The key parts to the definition are the phrases cumulative and excessive. During the time of the industrial revolution the common injury was considered repetitive motion, especially since many of the tasks were mechanical in nature, such as installing hundreds of wheels as part of an assembly line. With the rise of computers and their interfaces that require finer and more repetitive hand movements, as well as prolonged periods of sitting, there has been an increase in RSI's. Keller, Corbett, and Nichols (1998) writes of RSI's

> "In a BLS survey of nearly 90,000 work injuries resulting from repetitive motion, 12% were attributed to the repetitive motion of 'typing or key entry.' The number of workers with keyboard-related injuries can be expected to rise; already 60 million Americans nearly half the U.S. work force-use computers on a daily basis in the workplace or at home." (Keller, Corbett, & Nichols 1998, pp.9)

The issue for medical practitioners is that RSI's over time have a cumulative effect on bodily functions. As individuals repeat the same activity the body tries to compensate, leading to loss of blood flow and other detrimental effects. There is some debate among the medical community as to the extent to which computer based work causes RSI. The key distinction lies in the development of repetitive strain as opposed to immediate injury. Researchers have shown that office work combines repeated strain with physical positions, which increase the potential to develop RSI. MacKinnon and Novak (1997), in their article demonstrating RSI in the workplace, illustrate the cumulative damage of RSI. Their article shows how certain muscle types will change given exposure to stress and overexertion. They also demonstrate the pressure placed on the nerve endings, particularly in the hand and when the fingers are in a flexed position, as they are when a mouse is used. They write, "studies have confirmed that flexed or extended wrist positions, as well as, finger flexion, increase pressure within the carpal tunnel in patients with CTS." (MacKinnon & Novak, 1997, pp. 5). Mackinnon also addresses the three factors mentioned above that go into the development of RSI. Those factors being the amount of tissue damage, the individual's parameters (their physical makeup) and the psychosocial factors. These three pieces of the RSI triangle will become very important as we move forward.

The solution to RSI has widely been ergonomics. There are several articles discussing the role of ergonomics in libraries. Summer (1996) in the article *Ergonomics Programs and Activities in Research Libraries* illustrates the problem as being one that has developed as libraries have become more and more dependent on computers. Summer (1996) writes, "In short, in many institutions streamlined workflows, combined with reduced staff have resulted in staff members increasingly being tied to their computers with less offline work to mix into their daily routines." (Summer, 1996, pp. 86). Summer further demonstrates how libraries have developed "ergonomics" programs. These programs largely focused on proper office posture and physical exercise. Rooney (1994) also addresses the issue of the environment. Rooney highlights the ways in which the individual perceives the world around them and how ergonomics can help alleviate the ways in which humans interact with different office pieces.

Much of the literature presented was done in the later nineties. The reason much research on ergonomics occurs around this time, is this is when libraries were seeing an influx of computers into the workplace. These computers altered the work being conducted as well as the physical interactions. However, by examining ergonomics, the researchers are only focusing on two aspects of the RSI triangle, those being the level of tissue damage and the individual parameters. By incorporating ergonomics libraries can decrease the exposure to certain activity thus decreasing the level of tissue damage. They can also increase the individual's physical makeup through exercise. While exercise can improve mood, the work will continue to be generated in the employee's absence so ergonomics does not account for psychosocial needs. By examining another line of research being conducted in a different discipline around this time it is possible to create a new way of viewing the assessment of ergonomics in libraries. The additional discipline allows for the consideration of the psychosocial needs of the individual a term that can be understood as "flow".

The flow

The psychosocial balance of the work being performed can be investigated by focusing on the understanding of "flow". Miahly Csikszentmihalyi (1991) was the first psychologist to discuss flow or what the author calls the psychology of optimal experience. The notion of optimal experience is generated from what Csikszentmihalyi (1991) has termed the eight elements of enjoyment. There is an important caveat to note before we address the eight elements. Csikszentmihalyi demonstrates that these conditions apply regardless of age, sex, gender, race, etc. This is a key point when later a possible solution is discussed that can incorporate the elements of flow into improving staff productivity in ILL units.

The first element of enjoyment for Csikszentmihalyi (1991) is a challenging activity that requires skill. For this element, it is important to view the activity as challenging; thus when the process is overcome there is a sense of accomplishment. Second is the merging of action and awareness. The focus for this element is in the immersion of the activity that action becomes a spontaneous act connected to the activity being conducted. Third and fourth respectively is the task provides clear goals and immediate feedback. The more a staff member has a clear understanding of the goal of the activity the easier it is to conceptualize what needs to be done. Fifth is concentration on the task at hand. In this element, a lack of outside distraction leads to a certain level of focus that enhances the possibility for enjoyment. Sixth, the paradox of control, involves an understanding that loss of control does not lead to any "real" consequences. Csikszentmihalyi provides the example in the book of a chess match. If one were to lose a chess match, there are no real-world consequences. Thus, there is a paradox between the need for control of the game and a need for a lack of consequences. Seventh, the loss of self-consciousness, occurs in the moment when the person is so focused on the task at hand that they do not have any mental energy left over to focus on the self. Eighth, the transformation of time has little meaning. The person is so invested in the activity that time "flies" by and they would perform the task regardless of time investment because it is a reward end unto itself. The concept of psychological feelings of optimization applies in a larger context to the overall wellbeing of the individual.

These concepts can be applied on a local library level to help improve staff workflow and reduce repetitive strain injury. Csikszentmihalyi (1991) writes, "The more a job inherently resembles a game – with variety, appropriate and flexible challenges, clear goals, and immediate feedback – the more enjoyable it will be regardless of the worker's level of development." (Csikszentmihalyi, 1991, pp. 152). Effectively if the Interlibrary Loan unit or the library in general can utilize different gaming peripherals, it is possible to begin to develop the criteria in which the psychology of optimal experience or flow may develop. Using gaming peripherals, it is possible to decrease the attention needed on repetitive tasks and allow staff members to more easily achieve flow, a “state of effortless concentration and enjoyment” (Csikszentmihalyi. 1997). Csikszentmihalyi describes flow as “a particular kind of experience that is so engrossing that it becomes autotelic, that is, worth doing for its own sake even though it may have no consequence outside itself” (Csikszentmihalyi, 1999, p. 824). Nielsen and Cleal note that flow has been associated with “job satisfaction, enthusiasm, and contentment” (Nielsen & Cleal, 2010, P. 180).

Combining the literature on RSI, ergonomics and flow helps illustrate a new opportunity for libraries when considering ways to reduce employee stress, strain, workload and the subsequent psychological effects. This is in the use of gaming peripherals in libraries. Gaming peripherals have several advantages in helping alleviate the potential for RSI because they can address all three aspects of the RSI triangle. A gaming peripheral can help alleviate the amount of tissue damage by greatly reducing the numbers of clicks and wrist movements. Second, they can work in conjunction with an ergonomics program to help the individual improve their overall physical fitness. This is because most of these peripherals are customizable. The individual employee can alter the clicks and movement over time and not get stuck in a certain format. Finally, by reducing the number of actions taken, providing clearer objectives when accomplishing tasks and giving the employee greater control, an employee will have a better chance of achieving the psychosocial balance associated with flow. This can mean that, even though work amounts increase, the individual will be able to handle it since they are confident that the tools they possess can accomplish the task that they are responsible for. Texas Tech University Libraries Document Delivery department did just that and the results were quite promising.

Integration of gaming peripherals

Staff members at Texas Tech University Libraries’ Document Delivery department receive and process over one hundred thousand requests in a year (see Figure 1). Processing requests exposes staff members to repetitive stress and potential injury. Further, these staff members must remain efficient to process all requests in a timely manner. As a result, gaming peripherals were incorporated into the daily workflow at Texas Tech University Libraries to improve speed, accuracy, and efficiency while reducing repetitive stress.

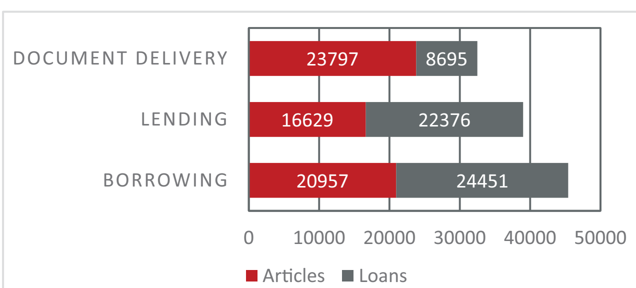

Figure 1. Total Requests for 2016 – 116,905.

The first gaming peripheral introduced into the workflow was the gaming mouse. Many different brands of gaming mice exist. Some of these include the Logitech G300s, the Logitech G600, and the Logitech G13, which were incorporated into the library workflow at Texas Tech University Libraries. Other options not utilized include the Razer, the SteelSeries and the Corsair. All these mice are well suited for the library workplace. The decision to use one over another reflects user preference rather than suitability. Gaming peripherals, like gaming mice, "have buttons that can handle hundreds of clicks a minute, and are outfitted with more sensitive sensors that allow you to aim your pointer with more precision, all but eliminating the tedious lifting and dragging that's sometimes required to get your pointer across a large screen" (Tang, 2014).

Although there are many brands and kinds of gaming mice, all have common features. The most important feature for the library workflow is the customizable buttons. The Logitech G300s, the first mouse incorporated into the library workflow, came equipped with eighteen customizable buttons, which allowed for repeated tasks to be mapped to a specific mouse button. Functions and commands most frequently used were mapped to the most easily accessible mouse buttons.

Borrowing specialists programed common tasks, such as copy and paste, to easily accessed buttons on the mouse (see figure 2). This allowed the specialists to quickly accomplish tasks without moving between the keyboard and mouse. In addition, other commonly used commands were mapped to the rest of the buttons, such as select all, enter and refresh. When a specialist opens a request, they can quickly and accurately paste information into a search, such as WorldCat, and process requests without the use of the keyboard.

The customization of these buttons allows workflow to proceed with fewer movements between keyboard and mouse and fewer keystrokes to complete common

Figure 2. Diagram of the Logitech mouse customization button layout.

Figure 3. Diagram of Staff member #1's Logitech G13 keypad customization button layout.

actions. This not only increases efficiency and speed, but also reduces the risk of repetitive stress injury. Staff members also increased the Dot Per Inch (DPI) of the gaming mice to increase responsiveness. DPI refers to how responsive a mouse is to movement. For example, a mouse with 1600 DPI will move 1600 pixels per inch. A higher DPI will be more responsive to movement. While a typical mouse has a DPI of 1600, a gaming mouse may have a DPI of 4000 or more and the responsiveness of the mouse will be programmable. The responsiveness of a mouse can help increase speed of processing and make tasks seem easier to perform with greater accuracy. Further, higher responsiveness means less arm movement, which can reduce mechanical stress for the staff member.

Finally, a gaming keypad was added with the mouse. The keypad is also constructed in an ergonomic shape. The keypad provided added functionality like the mice for more accurate copying and pasting (see Figures 3 and 4). Ergonomics, or arranging an environment to better fit its user, helps prevent potential injuries that can occur from repetitive stress and provides greater overall comfort. The gaming mice ergonomics have "been time-tested by devoted gamers" (Tang, 2014). Using gaming peripherals, which were designed with comfort and extended use in mind reduces the risk of repetitive strain injury and increases staff comfort.

Results

Incorporating gaming peripherals into the borrowing workflow improved the speed of processing. To determine whether there were any differences in the turnaround

Figure 4. Diagram of Staff member #2's Logitech G13 keypad customization button layout.

times after implementing gaming peripherals into the workflow, we examined transaction data for article requests from two separate time periods: 6/1/2015 to 12/1/2015 and 6/1/2016 to 12/1/2016. Article requests are typically more labor-intensive and complex than loan requests and therefore require more attention from staff members, which is why we chose to examine article data alone. We chose the time periods based on our busiest times of the year, when speed and accuracy is an even more important factor in our workflows. From the time periods we selected, we extracted a random sample of 800 requests each and determined the average turnaround time for each sample. From 6/1/2015 to 12/1/2015, our average turnaround time was 1.77 days. From 6/1/2016 to 12/1/2016, our turnaround time was 1.68 days. Overall, we found a difference of 2.1 hours between the samples.

Gamers value speed, efficiency, and accuracy, which are features that give an edge over opponents. In the ILL workplace, gaming peripherals can give staff an edge over increased workflow. By utilizing the features that give gamers an edge, ILL staff can process request with greater efficiency, thereby improving workflow. Incorporating gaming peripherals in the ILL workflow not only decreases the repetitive strain that can lead to workplace injury, but also increases efficiency, accuracy, and speed of processing, improving both the workflow itself and the conditions of working.

The incorporation of gaming peripherals likewise improves ergonomics and reduces repetitive stress. Library workflows often involve tasks that can cause repetitive stress, such as repetitive keyboard and mouse strikes. Like the ILL workflow, gaming involves repetitive motions and keystrokes, as well as potentially significant time spent using the same equipment, such as mice and keyboards. This stress can not only lead to discomfort, but to repetitive strain injury, such as carpal tunnel syndrome.

The example of above concerning interlibrary loan is applicable to other library tasks. Many tasks in technical services, circulation, and even digital services involve repeated copying and pasting of information. The ability to eliminate a mouse click or two greatly helps relieve both employee physical strain and psychological strain. Library workers are in an industry that is evolving to meet the needs of its users. Often this evolution requires doing more with less. By giving employees tools, they can use to both eliminate physical pain and increase psychological comfort of accomplishing a task helps not only improve the work environment, but it also fosters the conditions for employee satisfaction understood as the "flow".

Conclusions

Much of the literature involving ergonomics focuses on the relief of two of the three factors of repetitive strain injury. Those factors being the decrease in the amount of damage and accounting for the physical makeup of the individual by providing alternative postures. Repetitive strain injuries not only cause distress for staff members affected by the injury, but also cause departments loss of work and productivity. At a median of twenty-seven days, carpal tunnel syndrome causes the longest absences from work of any workplace injury (Bureau of Labor Statistics. 2004). According to the Occupational Safety and Health Administration (OSHA), repetitive strain injuries cost more than $20 billion a year in workers compensation and are the most common occupational health issue (OSHA). Thus, library management should pursue efforts to alleviate the stress caused by repetitive tasks in the library workflow. One way to both alleviate repetitive stress and increase efficiency is task optimization, "e.g. better design of tools and equipment and a better work lay-out". Tools frequently used in repetitive tasks include keyboards and mice, which can be optimized by better ergonomic design as well as by customization. However, very little of the research on repetitive stress focuses on the psychosocial aspects of the problem. By incorporating a gaming peripheral library staff can offer a way in which all three aspects of the repetitive strain injury triad can be addressed. The research above has shown that gaming peripherals have the potential to not only relieve the psychical pain, but they can also add to a better overall psychology in the hopes of working towards optimal enjoyment or flow. The focus on developing optimal employees allows for a different discussion of the role of ergonomics in the repetitive workload arena of libraries.

References

Bureau of Labor Statistics. (2004). *Lost-worktime injuries and illnesses: Characteristics adn resulting days away from work, 2002*. Web. 21 March 2017. https://www.bls.gov/news.release/archives/osh2_03252004.pdf

Byl, N. (2006). *Repetitive Strain Injuries*, 1381–1382.
Csikszentmihalyi, M. (1991). *Flow: The psychology of optimal experience*. New York: Harper Perennial Modern Classics.
Csikszentmihalyi, M. (1997). *Finding Flow*. Retrieved March 21, 2017, from https://www.psychologytoday.com/articles/199707/finding-flow
Csikszentmihalyi, M. (1999). If we are so rich, why aren't we happy? *American Psychologist*, *54*, 821–827. https://doi.org/10.1037/0003-066X.54.10.821.
Keller, K., Corbett, J., & Nichols, D. (1998). Repetitive strain injury in computer keyboard users: Pathomechanics and treatment principles in individual and group intervention. *Journal of Hand Therapy*, *11*(1), 9–26.
Mackinnon, & Novak (1997). Repetitive strain in the workplace. *Journal of Hand Surgery*, *22*(1), 2–18. https://doi.org/10.1016/S0363-5023(05)80174-1.
Nielsen, K., & Cleal, B. (2010). Predicting flow at work: Investigating the activities and job characteristics that predict flow states at work. *Journal of Occupational Health Psychology*, *15*(2), 180–190. Retrieved August 18, 2017. https://doi.org/10.1037/a0018893.
Rooney, James (1994). Ergonomics in academic libraries. *Library Management*, *15*(1), 26–35. https://doi.org/10.1108/01435129410049025.
Summer, S. (1996). Ergonomics programs and activities in research libraries. *Library Resources and Technical Services*, *40*(1), 84–92. https://doi.org/10.5860/lrts.40n1.84.
Tang, D. (2014). Gaming Mice for Nongamers. *Wall Street Journal* (Online), 1–2.